

LAPORAN PRAKTIKUM PEMROGRAMAN WEB MODUL 10:

Javascript Lanjutan

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Mata Kuliah : Pemrograman Web

2. Tujuan Praktikum

1. Memahami konsep dasar JavaScript sebagai bahasa pemrograman web.
2. Memahami fitur Modern JavaScript (ES6+) dan perannya sebagai fondasi React.
3. Mampu menggunakan sintaks ES6+ secara tepat dan konsisten.
4. Memahami konsep state, immutability, dan functional programming.
5. Mampu membangun aplikasi sederhana berbasis state menggunakan Vanilla JavaScript.

3. Alat dan Bahan

Perangkat lunak yang digunakan meliputi:

- Code editor (Visual Studio Code, Sublime Text, atau sejenisnya)
- Web browser

Bahan yang dibutuhkan:

- File HTML dasar
- File JavaScript (.js)

4. Langkah-langkah Praktikum

1. Persiapan: Membuat struktur dasar HTML (index.html) menggunakan template html:5.
2. Variabel & Scope: Membuat file let-const.js untuk menguji perbedaan let (bisa diubah) dan const (tetap/error jika diubah) serta membuktikan *block scope*.
3. Arrow Function: Membuat arrow-function.js untuk membandingkan sintaks fungsi biasa dengan *arrow function* yang lebih ringkas.
4. Destructuring & Spread: Menggunakan destructuring.js untuk mengekstrak nilai objek/array dan menggabungkan array tanpa mengubah data asli.
5. State & Immutability: Membuat immutability.js untuk simulasi pembaruan data yang tidak mengubah status lama, melainkan membuat data baru dengan spread operator.
6. Functional Programming: Mengimplementasikan Higher Order Functions (map, filter, reduce) pada file functional.js.
7. Asynchronous Fetch: Mengambil data dari API publik menggunakan fungsi fetch() dan menanganinya dengan .then() serta .catch() untuk error handling.
8. Modules: Memecah kode menjadi file utils.js dan main.js menggunakan fitur export dan import.

5. Hasil dan Dokumentasi Prongram

Index.html

```
<!doctype html>
<html lang="id">
<head>
<meta charset="UTF-8" />
<title>Vanilla JS Web Store</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

<link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Merr
iwe
ather:ital,wght@0,400;0,700;1,400&display=swap" rel="stylesheet">
<link rel="stylesheet" href="style.css" />

<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
<script type="module" src="app.js" defer></script>
</head>

<body>
<div id="app">
<div class="container">

<div class="listing-header">
<nav class="breadcrumb" aria-label="Breadcrumb">
<a href="#">
<svg xmlns="http://www.w3.org/2000/svg" fill="currentColor" viewBox="0 0 20 20">
<path d="m19.707 9.293-2-2-7-7a1 1 0 0 0-1.414 0l-7 7-2 2a1 1 0 0 0 1.414 1.414L2
10.414V18a2 2 0 0 0 2 2h3a1 1 0 0 0 1-1v-4a1 1 0 0 1 1-1h2a1 1 0 0 1 1 1v4a1 1 0
0 0 1 1h3a2
2 0 0 0 2-2v-7.586l.293.293a1 1 0 0 0 1.414-1.414Z" />
</svg>
Home
</a>
<svg style="color: #9CA3AF;" xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
stroke-
width="2" d="m9 5 7 7-7 7" />
```

```

</svg>
<a href="#">Products</a>
<svg style="color: #9CA3AF;" xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
stroke-
width="2" d="m9 5 7 7-7 7" />

</svg>
<span style="color: var(--text-muted);">Electronics</span>
</nav>

<div class="toolbar">
<h2 class="page-title">Electronics</h2>
<div class="toolbar-actions">
<button class="btn-secondary">

<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">
<path stroke="currentColor" stroke-linecap="round" stroke-width="2" d="M18.796
4H5.204a1 1 0 0 0-.753 1.659l5.302 6.058a1 1 0 0 1 .247.659v4.874a.5.5 0 0 0
.2.4l3 2.25a.5.5
0 0 0 .8-.4v-7.124a1 1 0 0 1 .247-.659l5.302-6.059c.566-.646.106-1.658-.753-
1.658Z" />
</svg>
Filters
</button>
<button class="btn-secondary">
<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
stroke-
width="2" d="M7 4v16M7 4l3 3M7 4 4 7m9-3h6l-6 6h6m-6.5 10 3.5-7 3.5 7M14 18h4" />

</svg>
Sort
</button>
</div>
</div>
</div>

<div id="product-list" class="product-grid"></div>

<div id="cart" class="cart-section"></div>

```

```

</div>
</div>

<script type="module" src="app.js"></script>
</body>
</html>

```

Style.css

```

:root {
  --primary: #0D9488;
  --primary-hover: #0F766E;
  --primary-light: #CCFBF1;
  --bg-color: #F9FAFB;
  --card-bg: #FFFFFF;
  --text-main: #111827;
  --text-muted: #6B7280;
  --border-color: #E5E7EB;
  --rating-star: #FACC15;
  --font-serif: 'Merriweather', serif;
  --font-sans: 'Inter', sans-serif;
  --radius: 8px;
}

* { box-sizing:
    border-box; margin: 0;
    padding: 0;
}

body {
  font-family: var(--font-sans);
  background: var(--bg-color);
  color: var(--text-main);
  line-height: 1.5;
  -webkit-font-smoothing: antialiased;
}

.container { max-width: 1280px; margin: 0 auto; padding: 32px 16px; }

.flex-row, .breadcrumb, .breadcrumb a, .toolbar, .toolbar-actions, .feature,
.rating-row, .card-
footer { display: flex; align-items: center; }

.btn-primary, .btn-secondary { display: inline-flex; align-items: center;
justify-content: center;

```

```
gap: 8px; border-radius: var(--radius); font-size: 0.875rem; font-weight: 500;
cursor: pointer;
transition: 0.2s; }
svg { width: 16px; height: 16px; }

.listing-header { margin-bottom: 32px; }
.page-title { font-family: var(--font-serif); font-size: 1.875rem; font-weight:
700; }
.toolbar { justify-content: space-between; flex-wrap: wrap; margin-top: 16px;
gap: 16px; }
.toolbar-actions { gap: 12px; }

.breadcrumb { gap: 8px; font-size: 0.875rem; color: var(--text-muted); margin-
bottom: 12px; }
.breadcrumb a { color: var(--text-muted); text-decoration: none; gap: 6px; }
.breadcrumb a:hover { color: var(--primary); }

.btn-secondary { background: white; border: 1px solid var(--border-color); color:
var(--text-main); padding: 8px 12px; }

.btn-secondary:hover { background: #F3F4F6; color: var(--primary); }

.btn-primary { background: var(--primary); color: white; border: none; padding:
10px 20px; }
.btn-primary:hover { background: var(--primary-hover); }
.btn-primary svg { width: 20px; height: 20px; }

.product-grid { display: grid; grid-template-columns: repeat(auto-fill,
minmax(250px, 1fr));
gap: 24px; margin-bottom: 32px; }

.product-card {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: var(--radius);
  overflow: hidden; display: flex;
  flex-direction: column;
  transition: box-shadow 0.2s;
  box-shadow: 0 1px 2px 0 rgba(0,0,0,0.05); }

.product-card:hover { box-shadow: 0 4px 6px -1px rgba(0,0,0,0.1); }

.card-image { height: 220px; padding: 24px; display: flex; justify-content:
center; background:
white; }
```

```

.card-image img { max-height: 100%; max-width: 100%; object-fit: contain; }
.card-content { padding: 24px; display: flex; flex-direction: column; flex: 1; }

.badge { background: var(--primary-light);
  color: var(--primary);
  font-size: 0.75rem;
  font-weight: 600;
  padding: 2px 10px;
  border-radius: 4px;
  width: fit-content;
  margin-bottom: 12px;
}

.card-title { font-size: 1.125rem; font-weight: 600; color: var(--text-main);
text-decoration:
none; margin-bottom: 8px;
line-height: 1.4; display:
-webkit-box;
-webkit-line-clamp: 2;
line-clamp: 2;
-webkit-box-orient: vertical;
overflow: hidden; }

.card-title:hover { text-decoration: underline; }

.rating-row { gap: 8px; margin-bottom: 16px; }
.stars { display: flex; color: var(--rating-star); }
.rating-score { font-size: 0.875rem; font-weight: 600; }
.rating-count, .feature { font-size: 0.875rem; color: var(--text-muted); }
.features-list { display: flex; gap: 12px; margin-bottom: 20px; }
.feature { gap: 6px; }
.card-footer { margin-top: auto; justify-content: space-between; }
.price { font-size: 1.5rem; font-weight: 800; }

#cart-sidebar { background: white; border-left: 1px solid var(--border-color);
padding: 24px;
width: 350px; position: fixed; right: 0; top: 0; bottom: 0; box-shadow: -4px 0
16px
rgba(0,0,0,0.05); overflow-y: auto; transform: translateX(100%); transition:
transform 0.3s; z-index: 50; display: flex; flex-direction: column; }

#cart-sidebar.open { transform: translateX(0); }
#cart-sidebar h2 { margin-bottom: 24px; font-family: var(--font-serif); }

```

```
.empty-msg { color: var(--text-muted); font-style: italic; }

.cart-list { flex: 1; overflow-y: auto; display: flex; flex-direction: column;
gap: 16px; margin-bottom: 24px; }

.cart-item { display: flex; align-items: center; gap: 12px; padding-bottom: 12px;
border-bottom: 1px solid var(--border-color); }

.cart-item img { width: 50px; height: 50px; object-fit: cover; border-radius:
4px; border: 1px
solid var(--border-color); }

.cart-details h4 { font-size: 0.875rem; font-weight: 600; margin-bottom: 4px;
display: -webkit-
box; -webkit-line-clamp: 1; line-clamp: 1; -webkit-box-orient: vertical;
overflow: hidden; }

.cart-details span { font-size: 0.875rem; color: var(--text-muted); }

.cart-footer-summary { border-top: 2px solid var(--border-color); padding-top:
20px; margin-top: auto; }

.total-row { display: flex; justify-content: space-between; align-items: center;
margin-bottom:
16px; font-weight: 700; font-size: 1.125rem; }
.total-price { color: var(--primary); }
#checkout-btn { width: 100%; }

.newsletter-section { margin-top: 64px; background: var(--card-bg); border-top:
1px solid var(--
border-color); padding: 48px 0; text-align: center; }
.newsletter-content { max-width: 500px; margin: 0 auto; }
.newsletter-content h3 { font-family: var(--font-serif); font-size: 1.5rem;
margin-bottom: 8px; }
.newsletter-content p { color: var(--text-muted); margin-bottom: 24px; }
.newsletter-content form { display: flex; gap: 8px; }

.newsletter-content input { flex: 1; padding: 10px; border: 1px solid var(--
border-color);
border-radius: var(--radius); }
#app { width: 100%; }
```


App.js

```
import { fetchProducts } from "../src/api/productApi.js";
import { getState, setState } from "../src/core/Store.js";
import { ProductList } from "../src/components/ProductList.js";
import { Cart } from "../src/components/Cart.js";

import { Newsletter } from "../src/components/Newsletter.js";

function addToCart(product) {
  const state = getState();

  setState({
    cart: [...state.cart, product],
  });

  render();
}

function checkout() {
  setState({ cart: [] });
  alert("Thank you for your purchase!");
  render();
}

function render() {
  const state = getState();

  ProductList(state.products, addToCart);
  Cart(state.cart, checkout);
}

async function init() {
  try {
    const data = await fetchProducts();
    setState({ products: data });
    render();
    Newsletter();
  } catch (error) {
    console.error("Failed to load products:", error);
  }
}

init();
```

src/api/product.js

```
export const fetchProducts = async () => {
  return new Promise((resolve) => {

    setTimeout(() => {
      resolve([
        {
          id: 1,
          title: "Mechanical Keyboard",
          price: 150,
          rating: { rate: 4.8, count: 230 },

          image: "https://images.unsplash.com/photo-1618384887929-16ec33fab9ef?auto=format&fit=crop&w=800&q=80"
        },
        {
          id: 2,
          title: "Gaming Mouse",
          price: 80,
          rating: { rate: 4.6, count: 150 },

          image: "https://images.unsplash.com/photo-1527814050087-3793815479db?auto=format&fit=crop&w=800&q=80"
        },
        {
          id: 3,
          title: "4K Monitor 24\"",
          price: 300,
          rating: { rate: 4.9, count: 85 },

          image: "https://images.unsplash.com/photo-1527443224154-c4a3942d3acf?auto=format&fit=crop&w=800&q=80"
        },
        {
          id: 4,
          title: "Premium Coffee Beans",
          price: 25,
          rating: { rate: 4.7, count: 500 },

          image: "https://images.unsplash.com/photo-1559056199-641a0ac8b55e?auto=format&fit=crop&w=800&q=80"
        },
        {
          id: 5,
```

```

title: "Travel Tumbler",
price: 35,
rating: { rate: 4.5, count: 120 },

image: "https://images.unsplash.com/photo-1578500494198-
246f612d3b3d?auto=format&fit=crop&w=800&q=80"
},
{
id: 6,
title: "Noise Cancelling Headphones",
price: 200,
rating: { rate: 4.8, count: 320 },

image: "https://images.unsplash.com/photo-1505740420928-
5e560c06d30e?auto=format&fit=crop&w=800&q=80"
},
]);
}, 800);
});
};

```

Src/components/Cart.js

```

export function Cart(cart, onCheckout) {
const container = document.getElementById("cart");
container.innerHTML = "<h2>Your Cart</h2>";

if (cart.length === 0) {
container.innerHTML += "<p class='empty-msg'>Cart is empty.</p>";
return;
}

const list = document.createElement("div");
list.className = "cart-list";

cart.map((item) => {
const div = document.createElement("div");
div.className = "cart-item";
div.innerHTML = `

<div class="cart-details">
<h4>${item.title}</h4>
<span>${item.price}</span>
</div>
`;

```

```

list.appendChild(div);
});

container.appendChild(list);

const total = cart.reduce((acc, item) => acc + (item.price || 0), 0);

const footer = document.createElement("div");
footer.className = "cart-footer-summary";
footer.innerHTML = `
<div class="total-row">
<span>Total:</span>
<span class="total-price">${total.toFixed(2)}</span>
</div>
<button class="btn-primary" id="checkout-btn">Buy Now</button>
`;

footer.querySelector("#checkout-btn").addEventListener("click", onCheckout);
container.appendChild(footer);
}

```

Src/components/newsletter.js

```

export function Newsletter() {
const $container = $('<div class="newsletter-section"></div>');
const $content = `(
<div class="newsletter-content">
<h3>Subscribe to our updates</h3>
<p>Get the latest tech news and offers.</p>
<form id="newsletter-form">
<input type="email" placeholder="Enter your email" required />
<button type="submit" class="btn-primary">Subscribe</button>
</form>
<div class="success-message" style="display:none; margin-top: 10px; color:
green;">

Thanks for subscribing!
</div>
</div>
`);

$content.append($content);
$("#app").append($container);

$content.find("form").on("submit", function (e) {

```

```

E.preventDefault();
const $form = $(this);

$form.slideUp(400, function() {
  // Show success message with fade in
  $content.find(".success-message").fadeIn();
});
});
}

```

Src/components/productlist.js

```

export function ProductList(products, onAdd) {
  const container = document.getElementById("product-list");
  container.innerHTML = "";

  products.map((product) => {
    const card = document.createElement("div");
    card.className = "product-card";

    const discount = Math.floor(Math.random() * 20) + 5;
    const rating = product.rating || { rate: 4.5, count: 100 };

    card.innerHTML = `
    <div class="card-image">
    
    </div>
    <div class="card-content">
    <span class="badge">Up to ${discount}% off</span>
    <a href="#" class="card-title" title="${product.title}">${product.title}</a>

    <div class="rating-row">
    <div class="stars">
    ${getStarSVG().repeat(5)}
    </div>
    <span class="rating-score">${rating.rate}</span>
    <span class="rating-count">(${rating.count})</span>
    </div>

    <ul class="features-list">
    <li class="feature">
    <svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

    <path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
    stroke-
    
```

```

width="2" d="M13 7h6l2 4m-8-4v8m0-8V6a1 1 0 0 0-1-1H4a1 1 0 0 0-1 1v9h2m8 0H9m4
0h2m4 0h2v-4m0 0h-5m3.5 5.5a2.5 2.5 0 1 1-5 0 2.5 2.5 0 0 1 5 0Zm-10 0a2.5 2.5 0
1 1-5 0
2.5 2.5 0 0 1 5 0Z" />
</svg>
Fast Delivery
</li>
</ul>

<div class="card-footer">
<span class="price">${product.price}</span>
<button class="add-btn btn-primary">
<svg xmlns="http://www.w3.org/2000/svg" fill="none" viewBox="0 0 24 24">

<path stroke="currentColor" stroke-linecap="round" stroke-linejoin="round"
stroke-
width="2" d="M4 4h1.5L8 16m0 0h8m-8 0a2 2 0 1 0 0 4 2 2 0 0 0 0-4Zm8 0a2 2 0 1 0
0 4 2 2

0 0 0 0-4Zm.75-3H7.5M11 7H6.312M17 4v6m-3-3h6" />
</svg>
Add
</button>
</div>
</div>
`;

card.querySelector(".add-btn").addEventListener("click", function () {
onAdd(product);
});

container.appendChild(card);
});
}

function getStarSVG() {
return `<svg xmlns="http://www.w3.org/2000/svg" fill="currentColor" viewBox="0 0
20 20">
<path d="M9.049 2.927c-.3-.921 1.603-.921 1.902 0l1.07 3.292a1 1 0
00.9569h3.462c.969 0 1.371 1.24588 1.811-2.8 2.034a1 1 0 00-.364 1.118l1.07
3.292c.3.921-
.755 1.688-1.54 1.118l-2.8-2.034a1 1 0 00-1.175 0l-2.8 2.034c-.784.57-1.838-.197-
1.539-

```

```
1.11811.07-3.292a1 1 0 00-.364-1.118L2.98 8.72c-.783-.57-.38-1.81.588-  
1.81h3.461a1 1 0  
00.951-.6911.07-3.292z"></path>  
</svg>`;
```

Src/core/store.js

```
import { deepClone } from "../utils.js";  
  
let state = {  
  products: [],  
  cart: [],  
};  
  
export const getState = () => deepClone(state);  
  
export const setState = (newState) => {  
  const nextState = { ...state, ...newState };  
  
  state = nextState;  
  
  console.log("[Store] State updated:", state);  
};
```

Src/core/utils.js

```
import { getState } from "../Store.js";  
  
export function deepClone(obj) {  
  if (obj === null || typeof obj !== "object") {  
    return obj;  
  }  
  
  if (obj instanceof Date) {  
    return new Date(obj.getTime());  
  }  
  
  if (Array.isArray(obj)) {  
    return obj.map((item) => deepClone(item));  
  }  
  
  const clonedObj = {};  
  for (const key in obj) {  
    if (Object.prototype.hasOwnProperty.call(obj, key)) {  
      clonedObj[key] = deepClone(obj[key]);  
    }  
  }  
}
```

```
}  
}  
  
return clonedObj;  
}
```

6. Analisis dan pembahasan

Analisis Variabel: Penggunaan `let` dan `const` lebih aman daripada `var` karena memiliki cakupan blok yang mencegah variabel "bocor" ke luar cakupan yang diinginkan.

Analisis Immutability: Dalam pengembangan modern (seperti React), data tidak boleh diubah langsung. Penggunaan spread operator memastikan data asli tetap utuh, yang memudahkan pelacakan perubahan *state*.

Analisis Asynchronous: Fungsi `fetch()` memungkinkan aplikasi tetap berjalan sambil menunggu data dari server, mencegah UI menjadi "beku" saat proses pemuatan data.

Analisis Modules: Dengan konsep modular, kode menjadi lebih terorganisir dan siap untuk diimplementasikan dalam arsitektur berbasis komponen.

7. Kesimpulan

Praktikum ini membuktikan bahwa fitur ES6+ bukan sekadar penyingkatan sintaks, melainkan fondasi penting untuk pengelolaan data yang aman dan efisien. Pemahaman tentang *immutability*, *functional programming*, dan *asynchronous fetch* sangat krusial sebelum melangkah ke pengembangan framework seperti React.

8. Lampiran

Link repository:

<https://github.com/AriqAhmadNayaka/WebPro4904/tree/MuhammadJimmiAlkadri-707012500012>